

SAGE: New Math for Next Generation AI Infrastructure



Most AI portfolios are powered by vastly inferior math.

SAGE MPW provides superior math — Training at $O(n)$. Inference at $O(n \log n)$.

01 AI breakthrough — it's just math

Quadratic cost drops to linear for training and near-linear for inference.

TRAINING

$$\cancel{O(n^2)} \rightarrow O(n)$$

INFERENCE

$$\cancel{O(n^2)} \rightarrow O(n \log n)$$

02 Market implications

a $\cancel{GW} \rightarrow kW$

Kilowatts instead of gigawatts

Data-center power and capital requirements fall by orders of magnitude — greener compute, with planetary-scale benefits.

“The best data center is the one you do not need to build.”

— Paul Charlton

b **Train + infer** on battery

Battery-powered devices do both in near real time

Personal and sovereign AI: own your inference instead of renting it from untrustworthy partners.

Smartphones

Smart infrastructure

03 Track record — who you are betting on

Paul Charlton Founder & tech lead

Inevitable Systems

Three decades building the hidden system layers that let computing cross its hardest thresholds: ubiquitous digital video, Java across devices, routers across cores, filesystems across operating systems, institutions across cloud, global-scale IoT.

30

years shipping foundational technology

Billions

of people rely on it daily

Apple QuickTime · Sun Java2D · Cisco IOS multicore · Intuit QuickBooks · SGI CXFS · HP storage · D.E. Shaw Cloud

— EVIDENCE

Digital Video — three decades of ubiquity

youtube.com/watch?v=41DLFG2ohzk&t=2774s

Team Leader — Paul Charlton

linkedin.com/in/techguru

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